按照XiangShanLab/xiangshan-course/docs/1-xiangshan-development-environment/Chapter_1_Overview_of_the_Xiangshan_Development_Environment.md at master · OpenXiangShan/XiangShanLab处的引导进行如下步骤。

克隆环境仓库、配置环境：

```
xpg@hw33:~/cie$ curl -I --connect-timeout 10 https://github.com
xpg@hw33:~/cie$ git clone https://github.com/OpenXiangShan/xs-env.git
Cloning into 'xs-env'...
remote: Enumerating objects: 1829, done.
remote: Counting objects: 100% (496/496), done.
remote: Compressing objects: 100% (171/171), done.
remote: Total 1829 (delta 375), reused 332 (delta 324), pack-reused 1333 (from 3)
Receiving objects: 100% (1829/1829), 8.94 MiB | 208.00 KiB/s, done.
Resolving deltas: 100% (991/991), done.
xpg@hw33:~/cie$ cd xs-env
xpg@hw33:~/cie/xs-env$ sudo -s ./setup-tools.sh
[sudo] password for xpg:
Hit:1 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal InRelease
Get:2 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates InRelease [128 kB]
Get:3 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-backports InRelease [128 kB]
Get:4 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates/main arm64 Packages [3,119 kB]
Get:5 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates/main Translation-en [601 kB]
Get:6 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates/main arm64 c-n-f Metadata [17.6 kB]
Get:7 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates/restricted arm64 Packages [63.8 kB]
Get:8 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates/restricted Translation-en [548 kB]
Get:9 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates/restricted arm64 c-n-f Metadata [428 B]
```

```
xpg@hw33:~/cie/xs-env$ source env.sh
SET XS_PROJECT_ROOT: /home/xpg/cie/xs-env
SET NOOP_HOME (XiangShan RTL Home): /home/xpg/cie/xs-env/XiangShan
SET NEMU_HOME: /home/xpg/cie/xs-env/NEMU
SET AM_HOME: /home/xpg/cie/xs-env/nexus-am
SET DRAMSIM3_HOME: /home/xpg/cie/xs-env/DRAMsim3
xpg@hw33:~/cie/xs-env$ source update-submodule.sh
error: pathspec 'origin/master' did not match any file(s) known to git
git submodule update --init
cd rocket-chip && git submodule update --init cde hardfloat
cd openLLC && git submodule update --init openNCB
Previous HEAD position was a48a11f2 perf(ame): add auto-vectorized floating-point MMACC paths (#1197)
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
remote: Enumerating objects: 208, done.
remote: Counting objects: 100% (185/185), done.
remote: Compressing objects: 100% (32/32), done.
```

编译香山仿真器：

```
xpg@hw33:~/cie/xs-env$ cd $NOOP_HOME
xpg@hw33:~/cie/xs-env/XiangShan$ make init
git submodule update --init
cd rocket-chip && git submodule update --init cde hardfloat
cd openLLC && git submodule update --init openNCB
```

```
xpg@hw33:~/cie/xs-env/XiangShan$ make emu CONFIG=MinimalConfig EMU_TRACE=1 -j60
mkdir -p build/rtl
Fri, 16 Oct 2026 06:50:07 +0000
time -avp -o ./build/time.log mill -i -Djvm-xmx=40G -Djvm-xss=256m xiangshan.test.runMain top.SimTop    \
       --target-dir build/rtl --config MinimalConfig --issue E.b --firtool-opt "--repl-seq-mem --repl-seq-mem-file=SimTop.sv.conf"           \
       --num-cores 1 --dump-fir --target systemverilog --split-verilog --firtool-opt "-O=release --disable-annotation-unknown --lowering-options=explicitBitcast,disallo
wLocalVariables,disallowPortDeclSharing,locationInfoStyle=none"  --enable-difftest --full-stacktrace
Preparing Java 21.0.12 runtime; this may take a minute or two ...
  [60/66] ============== xiangshan.test.runMain top.SimTop --target-dir build/rtl --config Min...eclSharing,locationInfoStyle=none --enable-difftest --full-stacktrace =
========
[build.sc-62/66] compile
[build.sc-62] [info] compiling 7 Scala sources to /home/xpg/cie/xs-env/XiangShan/out/mill-build/compile.dest/classes ...
[build.sc-62] [warn] /home/xpg/cie/xs-env/XiangShan/build.sc:273:88: constructor Locale in class Locale is deprecated (since 19)
[build.sc-62] [warn]                                DateTimeFormatter.ofPattern("MMM-dd-yyyy").withLocale(new Locale("en")))
[build.sc-62] [warn]                                                                       ^
[build.sc-62] [warn] /home/xpg/cie/xs-env/XiangShan/build.sc:280:97: constructor Locale in class Locale is deprecated (since 19)
```

修改hello程序，使之符合大赛的提交要求（由于没有说明需要哪个IP地址，这里用了校园网的出口公网IP）：

1. 第一阶段提交要求

1.1 环境部署验证：输出 hello xiangshan, I am xxx, IP address（xxx替换为团队或个人标识）

hello.c M

nexus-am > apps > hello > hello.c

```c
#include <klib.h>

int main()
{
    printf("hello xiangshan, I am xpg, IP address 111.42.148.39\n");
    return 0;
}
```

编译hello程序：

```
xpg@hw33:~/cie/xs-env/XiangShan$ cd $NOOP_HOME
xpg@hw33:~/cie/xs-env/XiangShan$ cd "$AM_HOME/apps/hello"
xpg@hw33:~/cie/xs-env/nexus-am/apps/hello$ make ARCH=riscv64-xs
 # Building hello [riscv64-xs] with AM_HOME {/home/xpg/cie/xs-env/nexus-am}
 + CC hello.c
 # Building lib-am [riscv64-xs]
 + AS src/nemu/common/mainargs.S
 + AR -> build/am-riscv64-xs.a
 # Building lib-klib [riscv64-xs]
 # Creating binary image [riscv64-xs]
 + LD -> build/hello-riscv64-xs.elf
 + OBJCOPY -> build/hello-riscv64-xs.bin
xpg@hw33:~/cie/xs-env/nexus-am/apps/hello$ ./build/emu -i $AM_HOME/apps/hello/build/hello-riscv64-xs.bin --no-diff > hello.log
 bash: ./build/emu: No such file or directory
xpg@hw33:~/cie/xs-env/nexus-am/apps/hello$ cd $NOOP_HOME
xpg@hw33:~/cie/xs-env/XiangShan$ ./build/emu -i $AM_HOME/apps/hello/build/hello-riscv64-xs.bin --no-diff > hello.log
```

运行hello程序，得到目标输出：

```
xpg@hw33:~/cie/xs-env$ cd $NOOP_HOME
xpg@hw33:~/cie/xs-env/XiangShan$ cd $NOOP_HOME
xpg@hw33:~/cie/xs-env/XiangShan$ ./build/emu -i $AM_HOME/apps/hello/build/hello-riscv64-xs.bin --no-diff
 emu compiled at Oct 16 2026, 07:36:36
 Using simulated 32768B flash
 Using simulated 8386560MB RAM
 The image is /home/xpg/cie/xs-env/nexus-am/apps/hello/build/hello-riscv64-xs.bin
 hello xiangshan, I am xpg, IP address 111.42.148.39
```

编写含向量加法的vadd程序，编译并运行，同时开启波形保存：

E: > Studying > RISC-V > cie > 第一阶段 > 261016 > vadd > C main.c

```c
#include <klib.h>
#include <stdint.h>

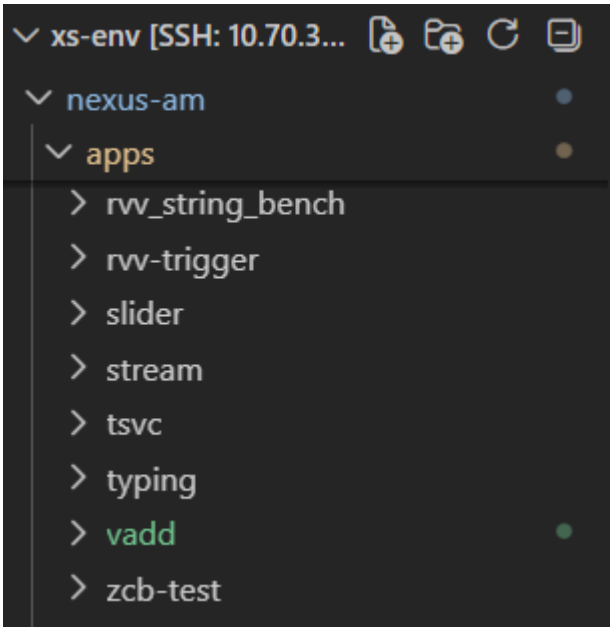
extern uintptr_t vadd_once(const uint8_t *a, const uint8_t *b, uint8_t *result);
extern const char vadd_target[];

int main(void) {
  static const uint8_t a[4] __attribute__((aligned(4))) = {0x2c, 0x2d, 0xba, 0xba};
  static const uint8_t b[4] __attribute__((aligned(4))) = {0x23, 0x25, 0xfe, 0xca};
  static uint8_t result[4] __attribute__((aligned(4))) = {0xff, 0xff, 0xff, 0xff};

  printf("vadd.vv test: SEW=8, LMUL=1, requested VL=2\n");
  printf("vadd_target PC: %p\n", (void *)vadd_target);
  uintptr_t vl = vadd_once(a, b, result);

  /* Check active elements and that the store leaves the other bytes intact. */
  if (vl != 2 || result[0] != 0x4f || result[1] != 0x52 ||
      result[2] != 0xff || result[3] != 0xff) {
    printf("FAIL: VL=%d, result=%x %x %x %x\n", (int)vl,
           result[0], result[1], result[2], result[3]);
    return 1;
  }
  printf("PASS: VL=2, result=4f 52 ff ff\n");
  return 0;
}

```



```
xpg@hw33:~/cie/xs-env/XiangShan$ cd "$AM_HOME/apps/vadd"
xpg@hw33:~/cie/xs-env/nexus-am/apps/vadd$ make ARCH=riscv64-xs disasm
# Building vadd [riscv64-xs] with AM_HOME {/home/xpg/cie/xs-env/nexus-am}
# Building lib-am [riscv64-xs]
+ AS src/nemu/common/mainargs.S
+ AR -> build/am-riscv64-xs.a
# Building lib-klib [riscv64-xs]
# Creating binary image [riscv64-xs]
+ LD -> build/vadd-riscv64-xs.elf
+ OBJCOPY -> build/vadd-riscv64-xs.bin
/usr/local/llvm/bin/llvm-objdump -d /home/xpg/cie/xs-env/nexus-am/apps/vadd/build/vadd-riscv64-xs.elf > /home/xpg/cie/xs-env/nexus
-am/apps/vadd/build/vadd-riscv64-xs.rvv.txt
RVV disassembly: /home/xpg/cie/xs-env/nexus-am/apps/vadd/build/vadd-riscv64-xs.rvv.txt (search for <vadd_target>)
xpg@hw33:~/cie/xs-env/nexus-am/apps/vadd$ cd "$NOOP_HOME"
xpg@hw33:~/cie/xs-env/XiangShan$ ./build/emu \
>   -i "$AM_HOME/apps/vadd/build/vadd-riscv64-xs.bin" \
>   --no-diff --dump-wave \
>   --wave-path="$AM_HOME/apps/vadd/build/vadd.vcd" \
>   > "$AM_HOME/apps/vadd/build/wave-run.log" 2>&1
```

波形分析见“向量加法指令vadd.vv执行过程分析.md“文档